AHMAD MSC(MATH)
Areeb2113@gmail.com

Q.No 1 Prove that if $f$ is continuous on a finite closed Interval $[a,b]$ then $f$ is bounded on $[a,b]$ (5)

Q.No 2 suppose that $f$ is monotonic on $(a,b)$ and define $\alpha = \inf_{a<x<b} f(x)$ If $f$ is nondecreasing then prove that $\lim_{x \to a+} f(x) = \alpha$. (5).

Q Define a Continuously differentiable function

Q. If Evaluate $\lim_{x \to \infty} \dfrac{\ln \sin x}{\ln x}$.

Q Show that the function $g(x) = x^2$ is uniformly continuous on $[-r, r]$ for any finite $r$.

Q. State Generalize mean value theorem

Q1

**Theorem**: If $f$ is continuous on a finite closed interval $[a, b]$, then $f$ is bounded on $[a, b]$.

**Proof**: Suppose that $t \in [a, b]$. Since $f$ is continuous at $t$, there is an open interval $I_t$ containing $t$ such that

$$|f(x) - f(t)| < 1 \quad \text{if} \quad x \in I_t \cap [a, b].$$

The collection $\mathcal{H} = \{I_t : a \leq t \leq b\}$ is an open covering of $[a, b]$.

Since $[a, b]$ is compact, the Heine-Borel theorem implies that there are finitely many points $t_1, t_2, \ldots, t_n$ such that the intervals $I_{t_1}, I_{t_2}, \ldots, I_{t_n}$ cover $[a, b]$.

According to continuity of the function and with $t = t_i$,

$$|f(x) - f(t_i)| < 1 \quad \text{if} \quad x \in I_{t_i} \cap [a, b].$$



Therefore,

$$\begin{aligned} |f(x)| &= |(f(x) - f(t_i)) + f(t_i)| \leq |f(x) - f(t_i)| + |f(t_i)| \\ &\leq 1 + |f(t_i)| \quad \text{if} \quad x \in I_{t_i} \cap [a, b]. \end{aligned}$$

Let

$$M = 1 + \max_{1 \leq i \leq n} |f(t_i)|.$$

Since $[a, b] \subset \bigcup_{i=1}^{n} (I_{t_i} \cap [a, b])$, and we have $|f(x)| \leq M$ if $x \in [a, b]$.

Q2

**Theorem:** Suppose that $f$ is monotonic on $(a, b)$ and define

$$\alpha = \inf_{a<x<b} f(x) \quad \text{and} \quad \beta = \sup_{a<x<b} f(x).$$

1. If $f$ is nondecreasing, then $\lim_{x \to a+} f(x) = \alpha$ and $\lim_{x \to b-} f(x) = \beta$.
2. If $f$ is nonincreasing, then $\lim_{x \to a+} f(x) = \beta$ and $\lim_{x \to b-} f(x) = \alpha$. (Here $a+ = -\infty$ if $a = -\infty$ and $b- = \infty$ if $b = \infty$.)

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3. If $a < x_0 < b$, then $\lim_{x \to x_{0+}} f(x)$ and $\lim_{x \to x_{0-}} f(x)$ exist and are finite; moreover,

$$\lim_{x \to x_{0-}} f(x) \le f(x_0) \le \lim_{x \to x_{0+}} f(x)$$

if $f$ is nondecreasing, and

$$\lim_{x \to x_{0-}} f(x) \ge f(x_0) \ge \lim_{x \to x_{0+}} f(x)$$

if $f$ is nonincreasing.

**Proof:** We first show that $\lim_{x \to a+} f(x) = \alpha$.

If $M > \alpha$, there is an $x_0$ in $(a, b)$ such that $f(x_0) < M$. Since $f$ is nondecreasing, $f(x) < M$ if $a < x < x_0$. Therefore, if $\alpha = -\infty$, then $\lim_{x \to a+} f(x) = -\infty$.

If $\alpha > -\infty$, let $M = \alpha + \varepsilon$, where $\varepsilon > 0$.

Then $\alpha \le f(x) < \alpha + \varepsilon$, so

$$|f(x) - \alpha| < \varepsilon \quad \text{if} \quad a < x < x_0. \tag{3.14}$$

If $a = -\infty$, this implies that $f(-\infty) = \alpha$. If $a > -\infty$, let $\delta = x_0 - a$. Then (3.14) is equivalent to

$$|f(x) - \alpha| < \varepsilon \quad \text{if} \quad a < x < a + \delta,$$

which implies that $f(a+) = \alpha$.

We now show that $\lim_{x \to b-} f(x) = \beta$.

If $M < \beta$, there is an $x_0$ in $(a, b)$ such that $f(x_0) > M$.

Since $f$ is nondecreasing, $f(x) > M$ if $x_0 < x < b$. Therefore, if $\beta = \infty$, then $\lim_{x \to b-} f(x) = \infty$.

If $\beta < \infty$, let $M = \beta - \varepsilon$, where $\varepsilon > 0$. Then $\beta - \varepsilon < f(x) \le \beta$, so

$$|f(x) - \beta| < \varepsilon \quad \text{if} \quad x_0 < x < b. \tag{3.15}$$

If $b = \infty$, this implies that $f(\infty) = \beta$. If $b < \infty$, let $\delta = b - x_0$.

Then (3.15) is equivalent to

$$|f(x) - \beta| < \varepsilon \quad \text{if} \quad b - \delta < x < b,$$

which implies that $f(b-) = \beta$.

Q3

**Continuously differentiable** We say that $f$ is *continuously differentiable on* $[a, b]$ if $f$ is differentiable on $[a, b]$, $f'$ is continuous on $(a, b)$,

$$f'_+(a) = \lim_{x \to a+} f'(x)$$

, and

$$f'_-(b) = \lim_{x \to b-} f'(x).$$

Q5

**Example:** Show that the function $g(x) = x^2$ is uniformly continuous on $[-r, r]$ for any finite $r$.

**Solution:** To see this, note that

$$|g(x) - g(x') = |x^2 - (x')^2| = |x - x'|\,|x + x'| \le 2r|x - x'|,$$

so

$$|g(x) - g(x')| < \varepsilon \quad \text{if} \quad |x - x'| < \delta = \frac{\varepsilon}{2r} \quad \text{and} \quad -r \le x, x' \le r.$$

66

## 4.7 Generalized Mean Value Theorem

**Theorem:** If $f$ and $g$ are continuous on the closed interval $[a, b]$ and differentiable on the open interval $(a, b)$, then

$$[g(b) - g(a)]f'(c) = [f(b) - f(a)]g'(c)$$

for some $c$ in $(a, b)$.

**Proof:** The function